Given the code fragment:

```
// line n1
num = new int[10];
```

Which code fragment can be inserted at line n1 to enable the code to compile?

- **A.** int num[10];
- **B.** int[] num;
- **C.** new int num[];
- **D.** int[10] num;

The answer is: B

What is the meaning of "write once, run anywhere" in Java?

- **A.** It is a marketing statement because Java programs must be compiled for a specific platform in order to run.
- **B.** Java programs, after being compiled, can run on any platform or device even without a Java Virtual Machine.
- **C.** Java programs are designed to run only in web browsers and, thus, can run wherever there is a browser.
- **D.** Java programs can run on any Java Virtual Machine without being recompiled.

The Answer is: D

Given:

```
public static void main(String[] args) {
     int iterations = 100;

     while (count < iterations) {
          System.out.println("Iteration " + count);
          count++;
     }
}
```

What is the result?

- **A.** The program compiles and nothing is printed.
- **B.** Iteration plus an increasing number is printed 100 times.
- **C.** An error occurs during compilation.
- **D.** Iteration plus an increasing number is printed 99 times.

The answer is: B

Given the code fragment:

```
int[] arr1 = {1, 2, 3};
int[] arr2 = new int[2];
arr2[0] = 10;
System.out.print(arr1.length + " : " + arr2.length);
```

What is the result?

- **A.** 0 : 1
- **B.** 3 : 1
- **C.** 3 : 2
- **D.** 2 : 0

The answer is: C

Given the code fragment:

```
class Course {
      String name;
      static int count = 0;
      Course(String name) {
            this.name = name;
            count++;
      }
}
public class App {
      public static void main(String[] args) {
            Course c = new Course("Java Programming");
            // line n1
      }
}
```

Which code fragment, when inserted at line n1, enables the code to print Java Programming:1?

- **A.** System.out.println(name + ":" + count);
- **B.** System.out.println(Course.name + ":" + c.count);
- **C.** System.out.println(c.name + ":" + count);
- **D.** System.out.println(c.name + ":" + Course.count);

The answer is: D

Given the code fragment:

```
String s1 = "foo-bar";
String s2 = new String ("foo-bar");
System.out.print (s1.equals(s2) + " ");
System.out.print (s1 == s2);
System.out.print (" " + s1.compareTo (s2));
```

What is the result?

- **A.** true false 0
- **B.** false true -1
- **C.** false false -1
- **D.** true true 0

The answer is: B

Which statement is true about a mutator method?

- A. It returns mutated instance members.
- B. It can be used to assign data to instance members.
- C. It must be declared private.
- D. It replaces the default constructor.

The answer is: B

Given the code fragment:

```
int number = 1;
String s = null;
try {
    number = s.length();
    number += 2;
}
catch (RuntimeException e) {
    number += 4;
}
System.out.println (number);
```


What is the result?

- A. 3
- B. Nothing is printed.
- C. 1
- D. 5

The Answer is: D

Given:

```
public class App{
      int num;
      public int add(int x) {
            num = num + x;
            return num;
      }
      public void add(int x) {
            num = num + x;
      }
      public static void main(String[] args) {
            App obj = new App();
            obj.add (100);
            int ans = obj.add(100);
            System.out.println(obj.num);
      }
}
```

What is the result?

- **A.** 200
- **B.** 100
- **C.** 300
- **D.** A compilation error occurs.

The Answer is: D

Identify two class variables.

- ☐ **A.** private Measure cm;
- ☐ **B.** public int size = 10;
- ☐ **C.** public static int counter = 0;
- ☐ **D.** private static int numberOfSquares = 20;
- ☐ **E.** int scale = 35;

The answer is: C and D

Which statement is true about a mutator method?

- ○ **A.** It returns mutated instance members.
- ○ **B.** It can be used to assign data to instance members.
- ○ **C.** It must be declared private.
- ○ **D.** It replaces the default constructor.

The answer is: B

Given the code fragment:

```
int number = 1;
String s = null;
try {
    number = s.length();
    number += 2;
}
catch (RuntimeException e) {
    number += 4;
}
System.out.println (number);
```


What is the result?

- **A.** 3
- **B.** Nothing is printed.
- **C.** 1
- **D.** 5

The answer is: 5

Given:

```
public class App{
      int num;
      public int add(int x) {
            num = num + x;
            return num;
      }
      public void add(int x) {
            num = num + x;
      }
      public static void main(String[] args) {
            App obj = new App();
            obj.add (100);
            int ans = obj.add(100);
            System.out.println(obj.num);
      }
}
```

What is the result?

- **A.** 200
- **B.** 100
- **C.** 300
- **D.** A compilation error occurs.

The answer is: B

Identify two class variables.

- [ ] **A.** private Measure cm;
- [ ] **B.** public int size = 10;
- [ ] **C.** public static int counter = 0;
- [ ] **D.** private static int numberOfSquares = 20;
- [ ] **E.** int scale = 35;

The answer is: C and D

Given:

```
public class Test {
     static int var2 = 200;
     public static void print () {
          System.out.println (var2);
     }
     public void print(int var1) {                    // line n1
          System.out.println(var1);
          var2 = var2 + var1;                         // line n2
          print ();
     }
     public static void main(String[] args) {
          Test obj = new Test();
          obj.print(100);
     }
}
```

What is the result?

- **A.** A compilation error occurs at line n1.
- **B.** 100
  200
- **C.** 100
  100
- **D.** A compilation error occurs at line n2.

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day-9-tasks > Test2.java > Test2 > main(String[])

```java
public class Test2 {
    static int var2 = 200;
    public static void print(){
        System.out.println(var2);
    }
    public void print(int var1){
        System.out.println(var1);
        var2 = var2 +var1;
        print();
    }
    Run | Debug
    public static void main(String[] args) {
        Test2 obj = new Test2();
        obj.print(var1: 100);
       // print();
    }
}
```

PROBLEMS 9   OUTPUT   TERMINAL   ...   powershell - day-9-tasks

```
PS C:\java-training-certification\day-9-tasks> java Test2.java
PS C:\java-training-certification\day-9-tasks> java Test2.java
100
300
PS C:\java-training-certification\day-9-tasks>
```

main*   ⓧ 4 ⚠ 4 ⓘ 1   Java: Ready   UTF-8   CRLF   { } Java   Go Live

Given the code fragment:

```
List<String> names = new ArrayList<> ();
names.add("Julia");
names.add("Peter");
for (Iterator<String> itr = names.iterator (); itr.hasNext();)
    System.out.println(itr.next());
```

What is the result?

- **A.** Julia
  Peter
- **B.** Peter
  Julia
- **C.** A compilation error occurs.
- **D.** A runtime exception is thrown.

The answer is: A

Which statement is true about exception handling?

- **A.** All catch blocks must be ordered from general to most specific.
- **B.** All statements in a try block are executed, even if an exception occurs in the middle of the try block.
- **C.** At least one catch block must accompany a try statement.
- **D.** At least one statement in a try block must throw an exception.

The answer is: B

Given the code fragment:

```
List<String> names = new ArrayList<>();
names.add ("Joe1");
names.add ("Paul");
names.remove (0);
names.remove (0);
System.out.println (names.isEmpty());
names.add ("Joel");
names.add ("Paul");
names.clear();
System.out.println (names.isEmpty ());
```


What is the result?

- **A.** true
  true
- **B.** A runtime exception is thrown
- **C.** false
  false
- **D.** false
  true

The answer is: A

Given the code fragment:

```
System.out.println(10 == 10 && ! (5 != 5));
System.out.println(10 < 8 || 10 > 2);
```

What is the result?

- **A.** true
  true
- **B.** true
  false
- **C.** false
  false
- **D.** false
  true

The answer is: A